TMA4130/35 H25

Opg 2, 4, 6

Opg 2

Vi har punktene

(0,0) (1,2) (2,6) (3,12)

a) Vi skal finne interpolasjonspolynomet $p(x)$

Lagrange

Hvis vi har punkter $(x_0, y_0), \dots, (x_n, y_n)$
så er interpolasjonspolynomet

$$p(x) = \sum_{i=0}^{n} y_i \ell_i(x) \quad \text{der} \quad \ell_i(x) = \prod_{\substack{j=0 \\ j \neq i}}^{n} \frac{x - x_j}{x_i - x_j}$$

Navngi punktene

$x_0 = 0,\ y_0 = 0$
$x_1 = 1,\ y_1 = 2$
$x_2 = 2,\ y_2 = 6$
$x_3 = 3,\ y_3 = 12$

Sett opp Lagrange-formelen

$$p(x) = y_0 \ell_0(x) + y_1 \ell_1(x) + y_2 \ell_2(x) + y_3 \ell_3(x)$$

$$p(x) = 2\ell_1(x) + 6\ell_2(x) + 12\ell_3(x)$$

$\ell_0(x)$ faller bort, så trenger ikke finne den

Finn $\ell_1(x)$

$$\ell_1(x) = \frac{(x - x_0)}{(x_1 - x_0)} \cdot \frac{(x - x_2)}{(x_1 - x_2)} \cdot \frac{(x - x_3)}{(x_1 - x_3)}$$

$$\ell_1(x) = \frac{(x-0)}{(1-0)} \frac{(x-2)}{(1-2)} \frac{(x-3)}{(1-3)}$$

$$\ell_1(x) = x \frac{(x-2)}{-1} \frac{(x-3)}{-2}$$

$$\ell_1(x) = \frac{1}{2}(x-2)(x-3)$$

Finn $\ell_2(x)$

$$\ell_2(x) = \frac{(x-x_0)}{(x_2-x_0)}\frac{(x-x_1)}{(x_2-x_1)}\frac{(x-x_3)}{(x_2-x_3)}$$

$$\ell_2(x) = \frac{(x-0)}{(2-0)}\frac{(x-1)}{(2-1)}\frac{(x-3)}{(2-3)}$$

$$\ell_2(x) = \frac{x}{2}(x-1)\frac{x-3}{-1}$$

$$\ell_2(x) = -\frac{x}{2}(x-1)(x-3)$$

Finn $\ell_3(x)$

$$\ell_3(x) = \frac{(x-x_0)}{(x_3-x_0)}\frac{(x-x_1)}{(x_3-x_1)}\frac{(x-x_2)}{(x_3-x_2)}$$

$$\ell_3(x) = \frac{(x-0)}{(3-0)}\frac{(x-1)}{(3-1)}\frac{(x-2)}{(3-2)}$$

$$\ell_3(x) = \frac{x}{3}\frac{(x-1)}{2}(x-2)$$

$$\ell_3(x) = \frac{1}{6}x(x-1)(x-2)$$

Sett inn i polynomet

$$p(x) = 2\ell_1(x) + 6\ell_2(x) + 12\ell_3(x)$$

$$p(x) = 2\left(\frac{1}{2}x(x-2)(x-3)\right) + 6\left(-\frac{1}{2}x(x-1)(x-3)\right) + 12\left(\frac{1}{6}x(x-1)(x-2)\right)$$

$$p(x) = x(x-2)(x-3) - 3x(x-1)(x-3) + 2x(x-1)(x-2)$$

$$p(x) = x(x^2-5x+6) - 3x(x^2-4x+3) + 2x(x^2-3x+2)$$

$$p(x) = x^3 - 5x^2 + 6x - 3x^3 + 12x^2 - 9x + 2x^3 - 6x^2 + 4x$$

$$p(x) = x^3 - 3x^3 + 2x^3 - 5x^2 + 12x^2 - 6x^2 + 6x - 9x + 4x$$

$$p(x) = x^2 + x$$

b) $f(1{,}5) \approx p(1{,}5)$

$$p(1{,}5) = (1{,}5)^2 + 1{,}5$$

$$p(1{,}5) = 3{,}75$$

c) Siden x-verdiene er forskjellige finnes det ved interpolasjonsteoremet et unikt polynom av grad høyst 3 som interpolerer punktene.

Opg 4

a) $\int_{-1}^{1} f(s)\,ds \approx Q[f] = w_0 f(-1) + w_1 f(-\tfrac{1}{3}) + w_2 f(\tfrac{1}{3}) + w_3 f(1)$

Bruk $f(s) = 1$

$$\int_{-1}^{1} 1\,ds = [x]_{-1}^{1} = (1) - (-1) = 2$$

$$Q[1] = w_0 + w_1 + w_2 + w_3 \Rightarrow w_0 + w_1 + w_2 + w_3 = 2$$

Bruk $f(s) = s$

$$\int_{-1}^{1} s\,ds = \left[\tfrac{1}{2}s^2\right]_{-1}^{1} = \tfrac{1}{2}(1)^2 - \left(\tfrac{1}{2}(-1)^2\right) = \tfrac{1}{2} - \tfrac{1}{2} = 0$$

$$Q[s] = w_0(-1) + w_1(-\tfrac{1}{3}) + w_2(\tfrac{1}{3}) + w_3(1)$$

$$0 = -w_0 - \tfrac{1}{3}w_1 + \tfrac{1}{3}w_2 + w_3$$

Bruk $f(s) = s^2$

$$\int_{-1}^{1} s^2\,ds = \left(\tfrac{1}{3}s^3\right]_{-1}^{1} = \tfrac{1}{3}(1)^3 - \left(\tfrac{1}{3}(-1)^3\right) = \tfrac{1}{3} - \left(-\tfrac{1}{3}\right) = \tfrac{2}{3}$$

$$Q[s^2] = w_0(-1)^2 + w_1(-\tfrac{1}{3})^2 + w_2(\tfrac{1}{3})^2 + w_3(1)^2$$

$$\tfrac{2}{3} = w_0 + \tfrac{1}{9}w_1 + \tfrac{1}{9}w_2 + w_3$$

Bruk $f(s) = s^3$

$$\int_{-1}^{1} s^3\,ds = \left[\tfrac{1}{4}s^4\right)_{-1}^{1} = \tfrac{1}{4}(1)^4 - \left(\tfrac{1}{4}(-1)^4\right) = \tfrac{1}{4} - \tfrac{1}{4} = 0$$

$$Q[s^3] = w_0(-1)^3 + w_1(-\tfrac{1}{3})^3 + w_2(\tfrac{1}{3})^3 + w_3(1)^3$$

$$0 = -w_0 - \tfrac{1}{27}w_1 + \tfrac{1}{27}w_2 + w_3$$

Ligningssystem

$$\left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 2 \\ -1 & -1/3 & 1/3 & 1 & 0 \\ 1 & 1/9 & 1/9 & 1 & 2/3 \\ -1 & -1/27 & 1/27 & 1 & 0 \end{array}\right] \begin{matrix} R_2+R_1 \\ R_3-R_1 \\ R_4+R_1 \\ \sim \end{matrix} \left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 2 \\ 0 & 2/3 & 4/3 & 2 & 2 \\ 0 & -8/9 & -8/9 & 0 & -4/3 \\ 0 & 26/27 & 28/27 & 2 & 2 \end{array}\right] \begin{matrix} \tfrac{3}{2}R_2 \\ \sim \end{matrix} \left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 2 \\ 0 & 1 & 2 & 3 & 3 \\ 0 & -8/9 & -8/9 & 0 & -4/3 \\ 0 & 26/27 & 28/27 & 2 & 2 \end{array}\right]$$

$$\begin{matrix} R_3+\tfrac{8}{9}R_2 \\ R_4-\tfrac{26}{27}R_2 \\ \sim \end{matrix} \left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 2 \\ 0 & 1 & 2 & 3 & 3 \\ 0 & 0 & 8/9 & 8/3 & 4/3 \\ 0 & 0 & -8/9 & -8/9 & -8/9 \end{array}\right] \begin{matrix} R_3\cdot\tfrac{9}{8} \\ \sim \end{matrix} \left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 2 \\ 0 & 1 & 2 & 3 & 3 \\ 0 & 0 & 1 & 3 & 3/2 \\ 0 & 0 & -8/9 & -8/9 & -8/9 \end{array}\right] \begin{matrix} R_4+\tfrac{8}{9}R_3 \\ \sim \end{matrix} \left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 2 \\ 0 & 1 & 2 & 3 & 3 \\ 0 & 0 & 1 & 3 & 3/2 \\ 0 & 0 & 0 & 16/9 & 4/9 \end{array}\right]$$

$$\overset{\frac{9}{16}R_4}{\sim}\left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 2 \\ 0 & 1 & 2 & 3 & 3 \\ 0 & 0 & 1 & 3 & 3/2 \\ 0 & 0 & 0 & 1 & 1/4 \end{array}\right] \begin{array}{l} R_1-R_4 \\ R_2-3R_4 \\ R_3-3R_4 \\ \sim \end{array} \left[\begin{array}{cccc|c} 1 & 1 & 1 & 0 & 7/4 \\ 0 & 1 & 2 & 0 & 9/4 \\ 0 & 0 & 1 & 0 & 3/4 \\ 0 & 0 & 0 & 1 & 1/4 \end{array}\right] \begin{array}{l} R_1-R_3 \\ R_2-2R_3 \\ \sim \end{array} \left[\begin{array}{cccc|c} 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 3/4 \\ 0 & 0 & 1 & 0 & 3/4 \\ 0 & 0 & 0 & 1 & 1/4 \end{array}\right]$$

$$\begin{array}{l} R_1-R_2 \\ \sim \end{array} \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 1/4 \\ 0 & 1 & 0 & 0 & 3/4 \\ 0 & 0 & 1 & 0 & 3/4 \\ 0 & 0 & 0 & 1 & 1/4 \end{array}\right]$$

$$w_0 = w_3 = \frac{1}{4}, \quad w_1 = w_2 = \frac{3}{4}$$

$$w_0 + w_1 = \frac{1}{4} + \frac{3}{4} = 1 \checkmark \qquad 9w_0 + w_2 = 9\frac{1}{4} + \frac{3}{4} = 3 \checkmark$$

b) $\displaystyle\int_{x_k}^{x_k+h} f(x)\,dx = hf\left(x_k + \tfrac{1}{2}h\right) + \tfrac{1}{24}h^3 f''(\tilde{\xi}), \quad x_k < \tilde{\xi} < x_k + h$

Taylorutvikling rundt midtpunktet

$m = x_k + \frac{1}{2}h$

Midtpunktregel

$$\int_{x_k}^{x_k+h} f(x)\,dx \approx hf\left(x_k + \tfrac{1}{2}h\right)$$

Symmetri

Når vi integrerer rundt midtpunktet, forsvinner førsteordensleddet fordi det er odde

Definer midtpunktet

La $m = x_k + \frac{1}{2}h$

Da er intervallet $[x_k, x_k + h]$

Taylorutvikle f(x) rundt m

$$f(x) = f(m) + f'(m)(x-m) + \tfrac{1}{2}f''(\xi(x))(x-m)^2$$

der $\xi(x)$ ligger mellom x og m

Integrer begge sider

$$\int_{x_k}^{x_k+h} f(x)\,dx = \int_{x_k}^{x_k+h} \left[f(m) + f'(m)(x-m) + \tfrac{1}{2}f''(\xi(x))(x-m)^2\right]dx$$

$$\int_{x_k}^{x_k+h} f(x)\,dx = \int_{x_k}^{x_k+h} f(m)\,dx + \int_{x_k}^{x_k+h} f'(m)(x-m)\,dx + \tfrac{1}{2}\int_{x_k}^{x_k+h} f''(\xi(x))(x-m)^2\,dx$$

Første integral

$$\int_{x_k}^{x_k+h} f(m)\,dx = hf(m) = hf\left(x_k + \tfrac{1}{2}h\right)$$

Andre integral forsvinner

$$\int_{x_k}^{x_k+h} f'(m)(x-m)\,dx = f'(m)\int_{x_k}^{x_k+h}(x-m)\,dx = 0$$

Siden m er midtpunktet, er integralet av $x-m$ et symmetrisk intervall lik 0

Restleddet

$$\int_{x_k}^{x_k+h} f(x)\,dx = hf(m) + \frac{1}{2}\int_{x_k}^{x_k+h} f''(\xi(x))(x-m)^2\,dx$$

Ved middelverdisetningen for integraler kan vi skrive:

$$\int_{x_k}^{x_k+h} f''(\xi(x))(x-m)^2\,dx = f''(\tilde{\xi})\int_{x_k}^{x_k+h}(x-m)^2\,dx$$

Regn ut integralet

$$\int_{x_k}^{x_k+h}(x-m)^2\,dx$$

Siden m er midtpunktet setter vi $u = x-m$

Da går grensene fra $-\frac{h}{2}$ til $\frac{h}{2}$

$$\int_{x_k}^{x_k+h}(x-m)^2\,dx = \int_{-h/2}^{h/2} u^2\,du = \left[\frac{u^3}{3}\right]_{-h/2}^{h/2} = \frac{(h/2)^3}{3} - \frac{(-h/2)^3}{3} = \frac{h^3}{24} - \left(-\frac{h^3}{24}\right) = \frac{h^3}{12}$$

Sett inn

$$\frac{1}{2}f''(\tilde{\xi})\int_{x_k}^{x_k+h}(x-m)^2\,dx = \frac{1}{2}f''(\tilde{\xi})\frac{h^3}{12} = \frac{h^3}{24}f''(\tilde{\xi})$$

Siden $m = x_k + \frac{1}{2}h$ får vi:

$$\int_{x_k}^{x_k+h} f(x)\,dx = hf\left(x_k + \tfrac{1}{2}h\right) + \frac{h^3}{24}f''(\tilde{\xi})$$

Opg 6

a) Skriv tredjeordens ODE-en som et førsteordens system:

$$x'''(t)-2x''(t)-x'(t)+2x(t)=1-2t$$

med initialbetingelser $x(0)=1,\ x'(0)=1,\ x''(0)=0$

Høyereordens ODE → førsteordens system

Når vi har en høyereordens differensialligning, lager vi nye variabler: $u_1=x,\ u_2=x',\ u_3=x''$

Da blir alle ligningene førsteordens

Definer nye variabler

$$u_1(t)=x(t)$$
$$u_2(t)=x'(t)$$
$$u_3(t)=x''(t)$$

Deriver variablene

$$u_1'=x'=u_2$$
$$u_2'=x'=u_3$$
$$u_3=x''$$
$$u_3'=x'''$$

Løs den opprinnelige ligningen for x'''

$$x'''-2x''-x'+2x=1-2t$$

$$x'''=2x''+x'-2x+1-2t$$

$$u_3'=2u_3+u_2-2u_1+1-2t$$

Skriv systemet

$$\begin{cases} u_1'=u_2 \\ u_2'=u_3 \\ u_3'=-2u_1+u_2-2u_3+1-2t \end{cases}$$

Skriv på matriseform

$$\begin{bmatrix} u_1' \\ u_2' \\ u_3' \end{bmatrix}=\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -2 & 1 & 2 \end{bmatrix}\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}+\begin{bmatrix} 0 \\ 0 \\ 1-2t \end{bmatrix}$$

Initialbetingelser

Vi hadde $x(0)=1, x'(0)=1, x''(0)=0$

Siden $u_1=x, u_2=x', u_3=x''$:

$$\begin{matrix} u_1(0)=1 \\ u_2(0)=1 \\ u_3(0)=0 \end{matrix} \Rightarrow \begin{bmatrix} u_1(0) \\ u_2(0) \\ u_3(0) \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

Løsning

$$\begin{bmatrix} u_1' \\ u_2' \\ u_3' \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -2 & 1 & 2 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1-2t \end{bmatrix} \text{ med } \begin{bmatrix} u_1(0) \\ u_2(0) \\ u_3(0) \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

b) Vi har IVP:

$$x'(t) = 1 + t - x(t), \quad x(0) = 0$$

Vi skal bruke eksplisitt Euler

Standardform for ODE

$$x'(t) = f(t,x)$$

Eksplisitt Euler

$$x_{n+1} = x_n + hf(t_n, x_n) \text{ der } t_n = nh$$

i) Finn x_1, x_2, x_3

Vi har $x_0 = 0$ og $t_0 = 0$

$$x_{n+1} = x_n + hf(t_n, x_n)$$

$$x_{n+1} = x_n + h(1 + t_n - x_n)$$

$$x_1 = x_0 + h(1 + t_0 - x_0)$$
$$x_1 = 0 + h(1 + 0 - 0)$$
$$x_1 = h$$

$$x_2 = x_1 + h(1 + t_1 - x_1)$$
$$x_2 = h + h(1 + h - h)$$
$$x_2 = 2h$$

$$x_3 = x_2 + h(1 + t_2 - x_2)$$
$$x_3 = 2h + h(1 + 2h - 2h)$$
$$x_3 = 3h$$

ii) Finn uttrykk for x_n

$x_1 = h$, $x_2 = 2h$, $x_3 = 3h$

Vi gjetter $x_n = nh$

Men siden $t_n = nh$ skriver vi

$x_n = t_n$

iii) Gjett eksakt løsning

Siden Euler-verdiene gir $x_n = t_n$
gjetter vi $x(t) = t$

Sjekker:
$x(t) = t \Rightarrow x'(t) = 1$

$1 + t - x(t) \Rightarrow 1 + t - t = 1$

Initialbetingelsen $x(0) = 0$ stemmer også

$x(t) = t$

c) Runge-Kutta og trapesmetoden

Vi skal vise at Runge-Kutta metoden med Butcher tableau

$$\begin{array}{c|cc} 0 & 0 & 0 \\ 1 & \frac{1}{2} & \frac{1}{2} \\ \hline & \frac{1}{2} & \frac{1}{2} \end{array}$$

er det samme som trapesmetoden

$$x_{n+1} = x_n + \frac{h}{2}\left(f(t_n, x_n) + f(t_{n+1}, x_{n+1})\right)$$

Runge-Kutta fra Butcher tableau

$$K_i = f\left(t_n + c_i h, x_n + h\sum_j a_{ij} K_j\right)$$

og $x_{n+1} = x_n + h\sum_i b_i K_i$

Les av tableau

$$\begin{array}{c|cc} 0 & 0 & 0 \\ 1 & \frac{1}{2} & \frac{1}{2} \\ \hline & \frac{1}{2} & \frac{1}{2} \end{array}$$

$c_1 = 0$ $\quad a_{11} = 0$ $\quad a_{12} = 0$
$c_2 = 1$ $\quad a_{21} = \frac{1}{2}$ $\quad a_{22} = \frac{1}{2}$

$b_1 = \frac{1}{2}$ $\quad b_2 = \frac{1}{2}$

Finn K_1

$$K_1 = f(t_n + c_1 h, x_n + h(a_{11}K_1 + a_{12}K_2))$$

$$K_1 = f(t_n + 0h, x_n + h(0K_1 + 0K_2))$$

$$K_1 = f(t_n, x_n)$$

Finn K_2

$$K_2 = f(t_n + c_2 h, x_n + h(a_{21}K_1 + a_{22}K_2))$$

$$K_2 = f(t_n + 1 \cdot h, x_n + h(\tfrac{1}{2}K_1 + \tfrac{1}{2}K_2))$$

$$K_2 = f(t_{n+1}, x_n + \tfrac{h}{2}(K_1 + K_2))$$

Skriv oppdateringsformelen

$$x_{n+1} = x_n + h(b_1 K_1 + b_2 K_2)$$

$$x_{n+1} = x_n + h(\tfrac{1}{2}K_1 + \tfrac{1}{2}K_2)$$

$$x_{n+1} = x_n + \tfrac{h}{2}(K_1 + K_2)$$

Bruk oppdateringen i uttrykket for K_2

Vi har $x_{n+1} = x_n + \frac{h}{2}(K_1 + K_2)$ og $K_2 = f(t_{n+1}, x_n + \frac{h}{2}(K_1 + K_2))$

Siden $x_n + \frac{h}{2}(K_1 + K_2) = x_{n+1}$ får vi:

$$K_2 = f(t_{n+1}, x_{n+1})$$

Og vi hadde $K_1 = f(t_n, x_n)$

Sett dette inn i oppdateringsformelen

$$x_{n+1} = x_n + \tfrac{h}{2}(K_1 + K_2)$$

$$x_{n+1} = x_n + \tfrac{h}{2}(f(t_n, x_n) + f(t_{n+1}, x_{n+1}))$$

Dette er trapesmetoden